

Domain mapping: 0 = Child, 1 = College, 2 = Social. Within each column, the best result is bold and the second-best result is underlined.

Table 1. Out-of-domain accuracy (%) for seed 0. Results use IID model selection.

Algorithm	Child	College	Social	Avg.
ERM	<u>58.163</u>	50.358	55.000	54.507
Mixup	56.463	47.733	54.545	52.914
SagNet	52.381	46.301	57.727	52.136
Fish	53.741	<u>52.267</u>	<u>58.182</u>	<u>54.730</u>
SWAD	<u>58.163</u>	46.778	<u>59.091</u>	54.677
Arith	54.762	<u>53.938</u>	56.364	55.021
GGA	53.741	46.539	56.364	52.215
ALOFT-S	55.102	41.527	53.182	49.937
ALOFT-E	54.762	51.074	58.182	54.673
iDAG	58.503	50.597	61.364	56.821
iDAG-SWAD	58.503	54.654	56.364	<u>56.507</u>

Table 2. Out-of-domain accuracy (%) for seed 1. Results use IID model selection.

Algorithm	Child	College	Social	Avg.
ERM	50.680	55.609	54.545	53.611
Mixup	52.381	52.267	55.455	53.368
SagNet	51.701	50.358	61.818	<u>54.626</u>
Fish	51.701	49.642	54.545	51.963
SWAD	50.340	<u>56.086</u>	57.273	54.566
Arith	55.102	48.449	<u>57.727</u>	53.759
GGA	51.020	50.119	57.273	52.804
ALOFT-S	46.259	50.119	57.273	51.217
ALOFT-E	54.762	<u>56.802</u>	54.091	<u>55.218</u>
iDAG	<u>52.721</u>	57.041	55.909	55.224
iDAG-SWAD	50.340	42.243	<u>61.364</u>	51.316

Table 3. Out-of-domain accuracy (%) for seed 2. Results use IID model selection.

Algorithm	Child	College	Social	Avg.
ERM	<u>54.082</u>	60.143	58.636	<u>57.620</u>
Mixup	53.401	53.222	54.545	53.723
SagNet	52.721	51.790	56.364	53.625
Fish	53.741	49.642	60.000	54.461
SWAD	51.020	43.437	54.091	49.516
Arith	52.381	46.778	62.727	53.962
GGA	58.163	<u>56.086</u>	58.636	57.628
ALOFT-S	53.401	51.551	<u>62.273</u>	55.742
ALOFT-E	54.422	51.551	55.909	53.961
iDAG	49.320	52.029	61.818	54.389
iDAG-SWAD	55.102	55.609	55.909	55.540

Table 4. Out-of-domain accuracy (%) for seed 3. Results use IID model selection.

Algorithm	Child	College	Social	Avg.
ERM	54.082	52.029	54.091	53.400
Mixup	51.701	53.699	50.909	52.103
SagNet	54.762	50.119	44.091	49.657
Fish	<u>54.422</u>	56.086	55.909	<u>55.472</u>
SWAD	54.082	53.699	<u>56.818</u>	54.866
Arith	52.041	46.062	50.909	49.671
GGA	54.762	<u>55.847</u>	55.909	<u>55.506</u>
ALOFT-S	50.340	58.950	51.364	53.551
ALOFT-E	52.381	55.370	59.545	55.765
iDAG	50.680	56.802	51.364	52.949
iDAG-SWAD	51.701	50.597	<u>59.091</u>	53.796

Table 5. Out-of-domain accuracy (%) for seed 4. Results use IID model selection.

Algorithm	Child	College	Social	Avg.
ERM	<u>58.503</u>	<u>57.757</u>	60.000	58.753
Mixup	57.143	52.745	54.545	54.811
SagNet	56.463	48.210	60.000	54.891
Fish	56.122	57.041	57.273	56.812
SWAD	56.122	56.802	54.091	55.672
Arith	54.762	59.427	<u>60.909</u>	58.366
GGA	58.844	57.518	59.091	<u>58.484</u>
ALOFT-S	55.442	54.177	55.000	54.873
ALOFT-E	54.422	55.609	59.545	56.525
iDAG	54.762	48.449	63.636	55.616
iDAG-SWAD	54.762	56.086	54.545	55.131

Table 6. Mean out-of-domain accuracy (%) over five seeds. Results use IID model selection.

Algorithm	Child	College	Social	Avg.
ERM	<u>55.102</u>	55.179	56.454	55.578
Mixup	54.218	51.933	54.000	53.384
SagNet	53.606	49.356	56.000	52.987
Fish	53.945	52.936	57.182	54.688
SWAD	53.945	51.360	56.273	53.859
Arith	53.810	50.931	<u>57.727</u>	54.156
GGA	55.306	53.222	<u>57.455</u>	<u>55.328</u>
ALOFT-S	52.109	51.265	55.818	53.064
ALOFT-E	54.150	<u>54.081</u>	57.454	55.228
iDAG	53.197	52.984	58.818	55.000
iDAG-SWAD	54.082	51.838	<u>57.455</u>	54.458